Tilak Poudel

Software Engineer | M.Sc. Data Science Candidate | Lecturer & Technology Educator

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PROFESSIONAL SUMMARY

Software Engineer with 6+ years of experience building and maintaining web applications, REST APIs, integrations, and data-driven features. Skilled in TypeScript, Node.js, PHP, relational databases, automated testing, and reliable software delivery. Currently pursuing an M.Sc. in Data Science at Tribhuvan University, with interests in applying data analysis, machine learning, and AI to practical problems. Combines hands-on engineering experience with university teaching and mentoring, bringing a collaborative approach to building maintainable, user-focused software and helping teams use data effectively.

TECHNICAL SKILLS

- **Programming:** TypeScript, JavaScript, PHP, Python.
- **Backend and APIs:** Node.js, Express, Laravel, REST API development, API integration, authentication, input validation, error handling.
- **Databases:** PostgreSQL, MySQL, MongoDB, Prisma ORM, relational database design, SQL query optimization.
- **Software Delivery:** Unit testing, integration testing, debugging, defect resolution, Git, code review, Agile, requirements analysis.
- **Tools and Deployment:** Docker, CI/CD, feature flags, Linux, Jira, Trello.

SOFTWARE DEVELOPMENT EXPERIENCE

Full Stack Developer (Freelance) | Aval Consulting Group (UK Client) **Feb 2026 – Present**

- Delivered the **Civil Survey Management System** from requirements gathering through production deployment.
- Designed a normalized **PostgreSQL** schema with **Prisma ORM**, migrated legacy data, and implemented a maintainable Node.js/Express repository pattern in TypeScript.
- Integrated the **Google Gemini API** to automate professional report synthesis and text generation.

Software Engineer | DzangoLab **2021 – Apr, 2026**

- Coordinated with distributed teams in the **US, France, and Thailand** to clarify requirements, share progress, and deliver software across time zones.
- Collaborated with engineering teams from European organizations, including **ENGIE** (energy) and **CITEO** (packaging, paper recycling, and circular economy), to understand project needs and contribute software aligned with their goals.
- Worked with stakeholders to understand problems, document technical requirements, and manage development tasks and defects in **Jira and Trello**.
- Prepared **Architecture Decision Records (ADRs)** to document key design choices, trade-offs, and their rationale.
- Wrote **automated tests** for new features, covering expected behavior and relevant edge cases to help prevent regressions.
- Improved database query performance by **30%** through indexing and SQL query optimization.
- Implemented secure **mTLS authentication** across microservices and feature flags to support controlled, iterative releases.
- Built and maintained Docker-based **CI/CD** pipelines, improving deployment reliability and reducing downtime by **40%**.
- Delivered dashboards and data analytics for **Wenow's carbon-footprint tracking solutions**, helping translate sustainability data into useful insights.

Full Stack Developer | Bootwal R&D **2018 – 2021**

- Developed **full-stack applications and RESTful APIs** with authentication for research and development projects.
- Collaborated with stakeholders to plan and build the **Lumbini Province Employee Management System**, supporting activity tracking, promotion evaluation, and transfer recommendations.

ADDITIONAL EXPERIENCE

Part-Time Lecturer | Sahid Smarak College and Academia International College **2026 – Present**

- Teach UI/UX design, mobile programming, and machine learning through practical projects.

Workshop Facilitator | Sahid Smarak College, Kathmandu **June 2026**

- Designed and delivered a **7-day workshop** on ML-powered web application development; trained **20+ students** using Python, Pandas, Scikit-learn, and Flask.

Lecturer & Trainer | Nepathya College **2019 – 2025**

- Taught Algorithms, DBMS, and programming to **300+ undergraduate students**; designed practical curricula and mentored student projects.

PROJECTS

- **Lumbini Province Employee Management System:** Helped plan and develop a system to track employee activities, define promotion evaluation criteria, and recommend office transfers based on employee history.
- **Community Forest Management Web App:** Developed and successfully deployed a system for community forest operations, including stock, loans and debts, events, and billing. PHP (CodeIgniter), MySQL, jQuery, DataTables.
- **Civil Survey Management System:** Delivered a survey management application from requirements gathering through production deployment using Node.js, Express, TypeScript, PostgreSQL, and Prisma ORM.
- **M.Sc. Data Science Thesis Research (Ongoing):** Exploring an NLP approach to Sanskrit verse correction and meter identification, combining language-model techniques with domain-specific analysis. Tribhuvan University.

CERTIFICATIONS

- **Python and Statistics for Financial Analysis** | Coursera
- **Data Science Math Skills** | Duke University
- **Elements of AI** | University of Helsinki
- **Learning Linux Command Line** | LinkedIn Learning

VOLUNTEER WORK

Youth Preacher | ISKCON Nepal **2020 – Present**

- Conducted **50+ sessions** at schools, colleges, and corporate offices on time management, healthy habits, and personal growth, reaching **1,000+ participants**.
- Contributed to developing and maintaining a **website and mobile app with an AI chatbot** for His Holiness Jayapataka Swami.
- Developed and managed **IT-related work for Tilottama Gaushala**, including its website.

EDUCATION

- **Master of Science in Data Science (Ongoing)** | Tribhuvan University **2025 – Present**
- **Bachelor of Science in Computer Science and Information Technology (BSc. CSIT)** | Nepathya College, graduated with distinction **2015 – 2019**